

Feature: Support for IxOS REST API and web chassisUI from FlixOS on HTRex (KCOS based)

Glossary:

- FlixOS based systems:
 - (brief) Debian based system where apps are deployed as deb installers where apps/services either running on host or in incus containers
 - [FlixOS - IxOS Architecture Guild - Confluence](#)
- KCOS based systems:
 - (brief) Kubernetes based operating system where apps are deployed as helm charts running as microservices within containers
 - <http://api.is.keysight.com/kcos-docs-design/master/>

Requirements:

(R1) short term: preflight checks by Cobalt product that require a very small set of the REST API

- rest service that exposes:
 - GET /chassis
 - GET /cards
 - GET /ports
 - GET /rgroups
- service is accessed internally within the k8s cluster by cobalt through its k8s svc name
 - service is not exposed publicly (there would be no user benefit as it only has a subset of the API)

(R2) longer term: IxOS REST API and web chassisUI exposed publicly

What is IxOS REST API and web chassisUI

Runs only on FlixOS based systems built on waf framework: [WAF Overall Description - Web Apps - Confluence](#)

Available at: [https://\[hostname/ip\]/chassislab/chassisui](https://[hostname/ip]/chassislab/chassisui)

S1. Porting WAF based IxOS REST API and web chassisUI

Implications:

- web-platform
- packages:
 - //web/labs/chassis/
 - //web/packages/hoganmw/
 - //packages/IxOsWebUI/
 - ChassisUI
 - //packages/IxOsWebUI/RPFProxyClient/main/
- tightly coupled with WAF
- branching out from flxos based version
- build and packaging effort across all packages to get to a container that require branching out
- functionality wise:
 - brings a postgresql that is not needed
 - brings its own user management, session management, etc. instead of using the KCOS platform support
 - initially would need to disable any such feature
 - in the future would need to support these futures based on what exists in KCOS
 - auth:
 - initially would need to disable auth as waf can only run with auth enabled
 - in the future would need to provide auth integration with keycloak
 - VMT scans:
 - high probability of being vulnerable
 - waf is in the process of upgrading jetty and its deps in two stage process, where stage one would be completed in few months
 - waf would require including this project as well

Effort: high

S2. Have the IxOS REST API and web chassis UI without WAF

☒ Phase 0. Satisfy (R1)

- creating a new rest service: ixos-rest-service
 - needs to use rpf IxServer API
 - rpf provides bindings only in C#, c++, Java
 - why not using ixserverctl (command line tool):
 - does not provide the required interface
 - BlackDuck/VMT scans:

- the intention is to use latest jetty and latest packages
 - RPF dependencies seem very old, might require branching out and have them updated
 - //packages/rpFramework/main/src/JavaClientLib/clientlib/rpframework-clientlib.iml
 - waf-utils that RPF depends on
- it will be java based (jetty) such as in Phase 1 can leverage >80% of hoganmw implementation
- deployment:
 - have it as a standalone microservice by providing a helm chart deployment
 - why not include it in the ixServer container:
 - the ixServer container seems already too fat:
 - starts init as its main process that starts multiple services: sshd, ixServer, licensing, etc.
 - logs are stored on disk instead of stdout
 - greatly restricts the lifecycle of the service: be it from a blackduck/vmt scanning, pkg deps, future requirements: auth, storage, etc.
- helm chart deployment
 - namespace: ixos-rest, svc name: ixos-rest-service
- container
- package: ixos-restapi, [Files · main · ranger / ixos-rest-service · GitLab](#)
- Estimation: 8 man days, status: DONE

Phase 1. Satisfy (R2)

- ☐ web chassis UI: no frontend expertise, action item on: [Laurentiu Constantinescu](#) , [Rajesh Paramasivan](#)
 - need to consider integration with platform web UI: breadcrumb (navigational section), auth, settings menu, etc.
 - likely some portions of the UI would need to be redesigned to capture the next bullet entire list of functionalities
- ☐ ixos-restapi (base path: /chassis/api/v2)
 - extend to most of the list of supported IxOS REST API, the service of (R1) is extended to IxOS REST API
 - GET /
 - GET /\$options
 - GET /cardmodes
 - GET /cardmodes/\$options
 - GET /cardmodes/{id}
 - GET /cardmodes/{id}/\$options
 - POST /cardmodes/{id}/operations/setcurrent
 - GET /cards
 - POST /cards
 - GET /cards/{id}
 - DEL /cards/{id}
 - GET /cards/{id}/\$options
 - POST /cards/{id}/operations/hotswap
 - GET /chassis
 - GET /chassis/{id}
 - PATCH /chassis/{id}
 - GET /chassis/{id}/\$options
 - POST /chassis/{id}/operations/activatefeature
 - POST /chassis/{id}/operations/collectlogs
 - GET /chassis/{id}/operations/collectlogs/{asyncid}
 - GET /chassis/{id}/operations/collectlogs/{asyncid}/result
 - POST /chassis/{id}/operations/discovervirtualloadmodules
 - POST /chassis/{id}/operations/hotswapemptyslot
 - POST /chassis/{id}/operations/retrieveertlog
 - POST /chassis/{id}/operations/validatevirtualloadmodules
 - GET /operations
 - GET /operations/{asyncid}
 - GET /operations/{asyncid}/result
 - GET /ports
 - GET /ports/{id}
 - PATCH /ports/{id}
 - GET /ports/{id}/\$options
 - POST /ports/{id}/operations/clearownership
 - POST /ports/{id}/operations/reboot
 - POST /ports/{id}/operations/releaseownership
 - POST /ports/{id}/operations/resetfactorydefaults
 - POST /ports/{id}/operations/resetmodedefaults
 - POST /ports/{id}/operations/retrieveepromsettings
 - POST /ports/{id}/operations/setcoppermode
 - POST /ports/{id}/operations/setfibermode
 - POST /ports/{id}/operations/setsgmiimode
 - POST /ports/{id}/operations/simulatecableconnect
 - POST /ports/{id}/operations/simulatecabledisconnect
 - POST /ports/{id}/operations/simulatetxcableconnect
 - POST /ports/{id}/operations/simulatetxcabledisconnect
 - POST /ports/{id}/operations/takeownership
 - GET /portstats
 - GET /portstats/{id}
 - GET /rgroups
 - GET /rgroups/\$options
 - GET /rgroups/{id}

- GET /rgroups/{id}/\$options
- POST /rgroups/{id}/operations/setcurrent
- GET /transceivers
- GET /transceivers/{id}
- by leveraging the most of hoganmw implementation by decoupling it from waf
 - data model, tasks, caching built on top of the IxServer RPF API can move in a shared component
- Rough estimate: 8 man days, status: **DONE**

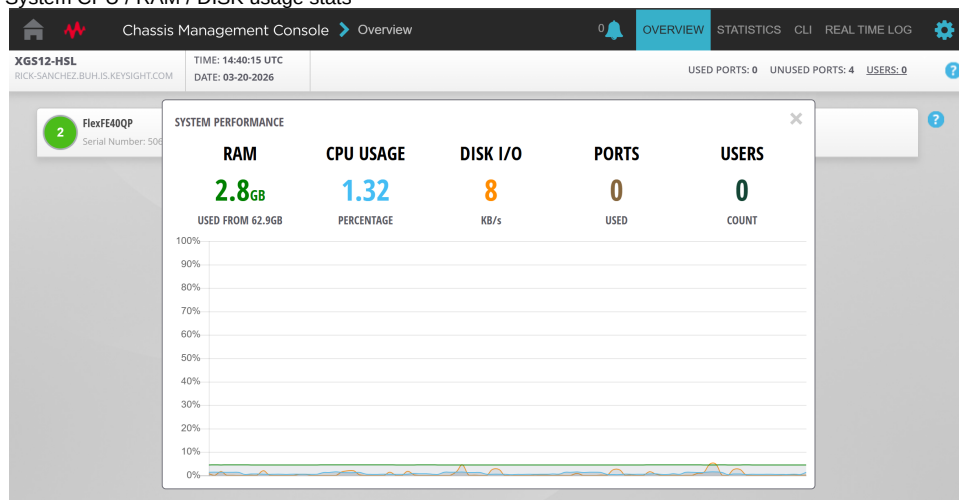
☐ the following categories need be accommodated for full support:

- ☐ 1. REST API endpoints from IxOSService from FlixOS dar are not applicable on KCOS as-is, e.g. <https://rick-sanchez.buh.is.keysight.com/chassis/swagger/index.html#/IxOSService>

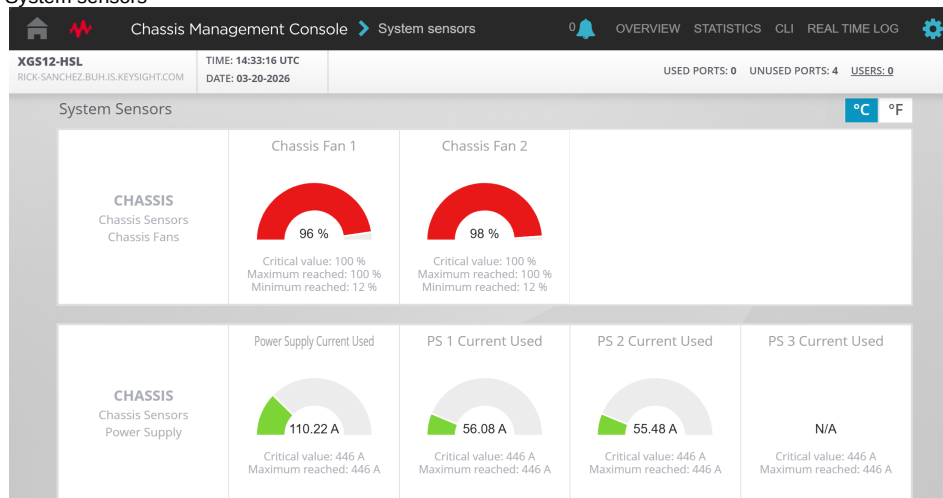
- POST /chassis/{id}/operations/cleanupdiskspace
- POST /chassis/{id}/operations/rebootchassis
- POST /chassis/{id}/operations/retrievechassistime
- POST /chassis/{id}/operations/retrievedhcpsettings
- POST /chassis/{id}/operations/setdhcpsettings
- POST /chassis/{id}/operations/setdhcpstatus
- GET /chassisstats
- GET /chassisstats/{id}
- GET /logs
- GET /logs/{id}
- POST /logs/{id}/operations/collect
- GET /logs/{id}/operations/collect/{asynclid}
- GET /logs/{id}/operations/collect/{asynclid}/result
- GET /perfcounters
- GET /sensors
- GET /sensors/{id}
- GET /services
- GET /services/{id}
- POST /services/{id}/operations/restart

- ☐ 2. How to support System level features (need to be redesigned and reimplemented for the KCOS platform)

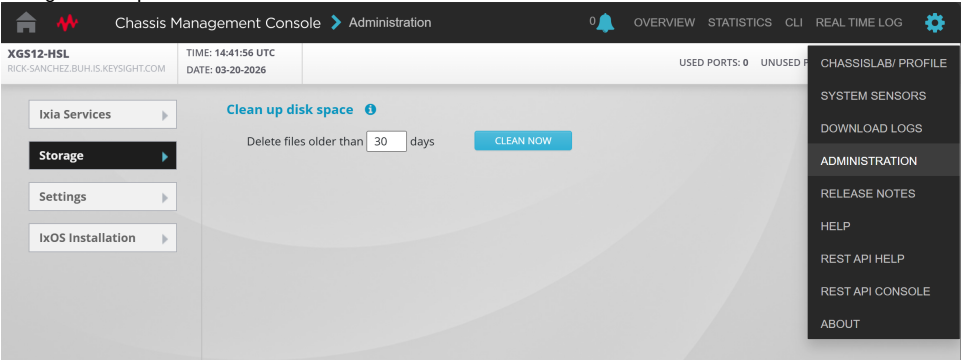
1. System CPU / RAM / DISK usage stats



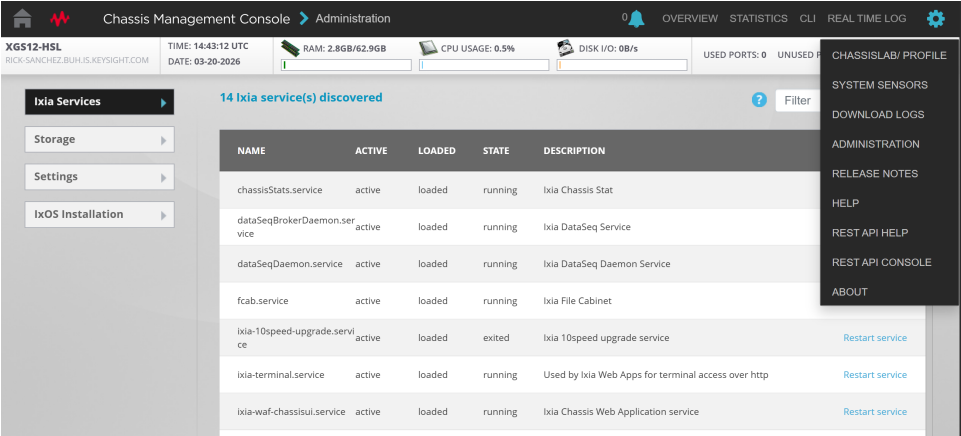
2. System sensors



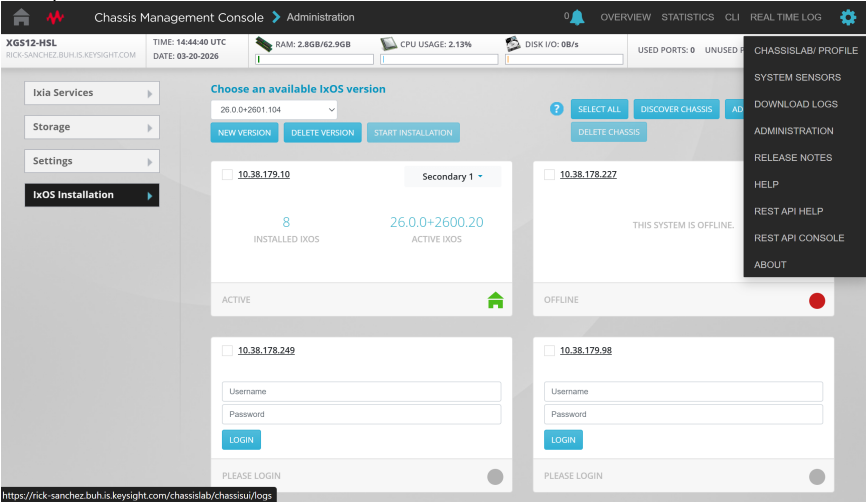
3. Storage clean up



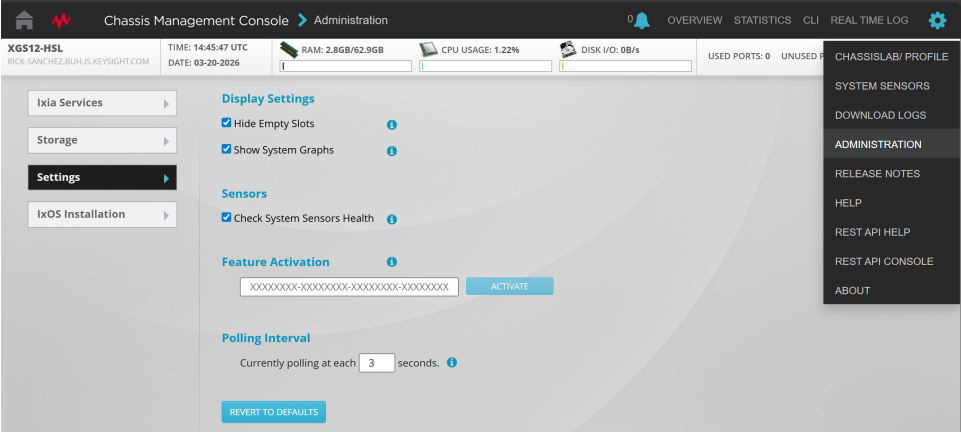
4. Ixia Services



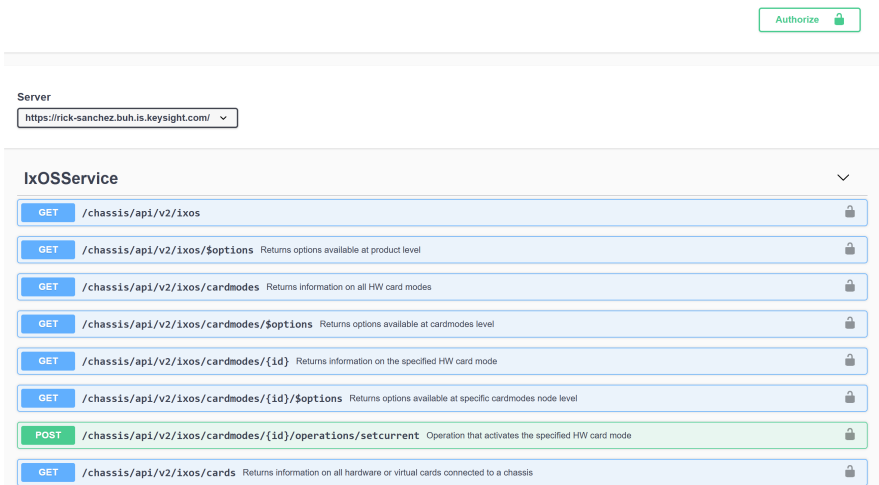
5. Bulk update / install IxoS



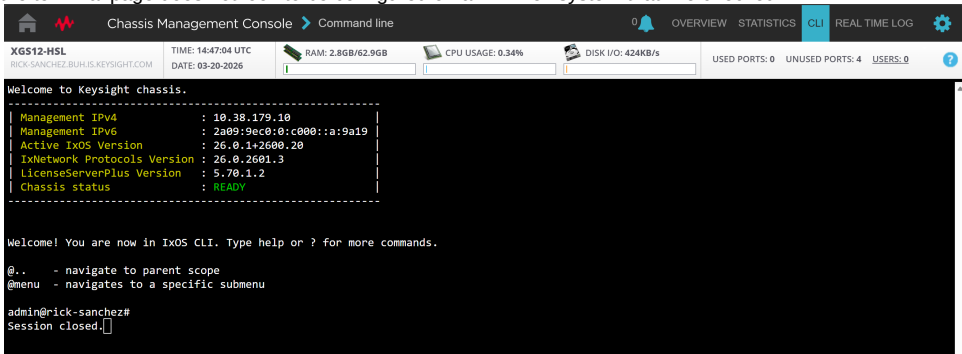
☐ 3. Verify IxoS settings (REST API present, need to confirm if it's 100% applicable to KCOS platform)



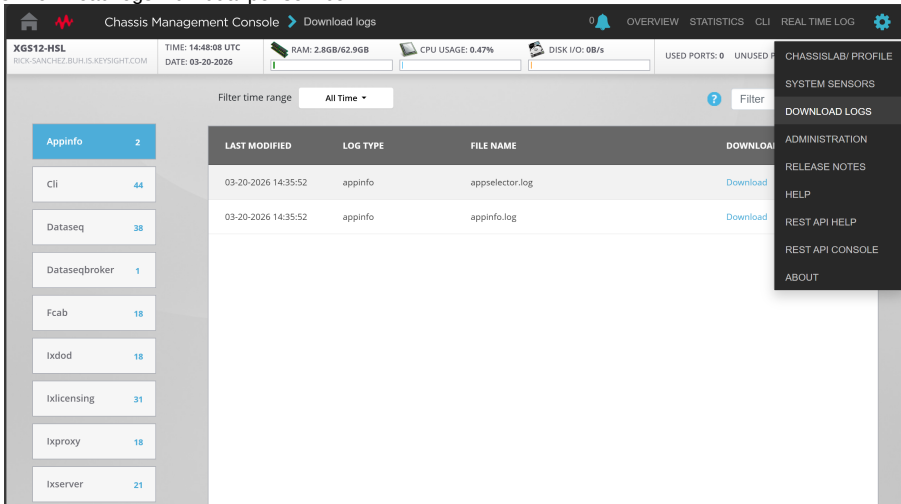
☐ Web API browser using Swagger, not present, e.g. from FlixOS <https://rick-sanchez.buh.is.keysight.com/chassis/swagger/index.html>



- ☐ 4. IxOS CLI in browser
- the KCOS dashboard includes a Terminal page that shows kcos-shell from where a user can do kcos enter ixos
 - the terminal page does not look to be configured on a HTTPREX system that I've checked



- ☐ 5. Download logs individual per service



- ☐ 6. Backed auth support/integration w/ kcos-ss0 (keycloak)